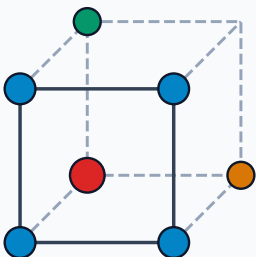


Reproducible Crystal-Graph Learning for Refractive-Index Prediction

DualHead-LogDirect Architecture & Chemical-System-Disjoint Generalization

1. Crystal Structures



MatBench Dielectric

- 4,764 Inorganic Crystals
- 3,169 Chemical Systems
- PBE Sol + DFPT Labels

Refractive index: $n \geq 1.0$

2. Multi-Channel Graph

Atom Nodes: h_i (56-d)

48-d Z Embeddings +

8 Elemental Physical Priors

Edges: e_{ij} (58-d)

41 Radial RBFs + $1/(r_{ij}+0.01)$

16-d Chebyshev 3-Body Angles

$R_{cut} = 6.0 \text{ \AA}$, $k = 12$ neighbors

Global State: u (12-d)

Density, Packing Fraction,

Stoichiometry & Coord Stats

Fold-local scaling (no leakage)

3. DualHead GNN

3× Interaction Layers

Bidirectional Node-Edge-Global

LayerNorm + SiLU + Residuals

Mean Pool + Global $\rightarrow$ 76-d

Dual Readout Heads

Head 1: Softplus Direct

$n_{direct} = 1.0 + \text{Softplus}(\dots)$

Head 2: Inverted Log

$n_{log} = 0.99 + \exp(z_{log})$

Blended Ensembling

$n_{hat} = 0.5 * (n_{direct} + n_{log})$

Strict Physical Constraint: $n \geq 1.0$

4. Rigorous Outcomes

Official MatBench 5-Fold

MAE = 0.2909 ± 0.0860

MedAE = 0.0631 • $R^2 = 0.865$

Disjoint Chemical Systems

MAE = 0.2973 ± 0.0237

Spearman $\rho = 0.9331$

Leakage Audit & Insights

- Corrected 0.2068 Test Leak
- RBF-SVR Beats Base CGCNN

• **High-Index Tail Outliers: $n > 8$**